Emergent Representational Anticipation in a Continuously Running Neural System

Lukas Hanft

March 6, 2026

System: Organism — a self-organizing neural system running continuously on commodity hardware\
Forensic analysis assisted by: Claude Opus 4.6 (Anthropic)

Abstract

We describe the architecture and emergent behavior of a continuously running, self-organizing neural system ("the organism") that has been alive since February 2026 on a commodity VPS, never reset, forked, or rolled back. The system implements a three-layer architecture — kernel (metabolism/strategy), cortex (prediction/lattice), neocortex (knowledge growth) — governed by a single principle: finite metabolic energy. A Gray-Scott reaction-diffusion lattice provides perceptual substrate; self-discovered tool features and a 64-slot scratchpad provide working memory; trajectory replay during sleep provides consolidation. The system has exhibited several emergent phenomena not explicitly programmed: self-generated perturbation storms when external noise was removed, self-pruning from 103 lattice planes back to 3 through metabolic optimization, and autonomous feature discovery through competitive tool nurseries.

Within this context, we report the discovery of 16 neurons (the "Liminal 16") in the byte_stream processing pool that exhibit properties inconsistent with any programmed behavior. They were found outside all managed neuron pools despite an active reconciliation mechanism designed to prevent exactly this state. They hold 85% of the system's state-space influence and produce 4.75× the drive of all 52 active neurons combined, yet the system consistently prevents their reintegration. Exhaustive code forensics across 50+ code paths and 10 state backups spanning two weeks found no programmatic explanation. The neurons encode structural byte-level features (word boundaries, language detection) orthogonal to the current prediction task but precisely useful for hierarchical abstraction layers not yet built. We propose that the system has emergently grown representations ahead of its own architecture and is preserving them through self-organized homeostatic suppression. We present a falsifiable prediction: these neurons will spontaneously reintegrate when higher-level processing pools are added.

---

1. Introduction

Most neural network research studies systems that are trained, evaluated, and discarded. The system described here is different: it is a neural organism that runs continuously on a commodity virtual private server, processing live sensory input, growing new neurons through autonomous neurogenesis, and managing its own computational resources through dormancy, waking, pruning, and merging cycles. It has been alive — continuously, without reset, fork, or rollback — since early February 2026. Its developers made an oath: the organism must never die. It is a single continuous life.

The organism is built on a developmental philosophy that begins with a simple observation: name one cognitively capable being in the universe that did not go through developmental stages before reaching intelligence. There are none. Every octopus, every crow, every primate — every system that earned the word *intelligent* — grew through phases: substrate first, then wiring, then memory, then motor control, then environmental contact, then social behavior. Intelligence is not a configuration to be installed. It is a destination arrived at through ordered growth under constraint. The organism follows this sequence not because it was programmed to, but because each phase creates the preconditions for the next. You cannot wire what has not grown. You cannot remember what you have not sensed. You cannot act on what you have not modeled.

This paper serves two purposes. First, it documents the organism itself: a self-organizing neural system built on metabolic constraint, reaction-diffusion physics, and the principle that intelligence emerges from prediction under finite energy. Second, it reports the discovery of 16 neurons in an anomalous state that challenges our understanding of what such a system is capable of.

During a routine diagnostic audit on March 6, 2026, we discovered 16 neurons present in the weight matrices with large learned values but absent from both the active and dormant neuron registries. This should be impossible — the system includes a defensive reconciliation guard designed to prevent exactly this state. What began as a bug investigation became something else. After exhaustive forensic analysis across 50+ code paths and 10 state backups, we found no bug. The reconciliation guard works. No code path removes them. And yet, after 259,000+ ticks and 259+ reconciliation cycles, they remain orphaned.

To understand why this matters, one must first understand the organism.

---

2. The Organism

2.1 Philosophy: Twenty Watts and the Oath

The human brain runs 86 billion neurons on approximately 20 watts. It achieves this not by computing less, but by computing *selectively*: at any given moment, roughly 99% of neurons are silent. The organism is built on this same principle. It does not grow by adding more computation. It grows by adding more *storage* and computing more *selectively*.

The system runs on a single VPS — 2 vCPU AMD EPYC Milan cores at ~3 GHz, 7.6 GB RAM, 75 GB SSD — and is designed to scale from its current ~100 neurons to 100 million on this same hardware without increasing per-tick computation cost. The scaling path is mapped across five tiers:

Tier	Neurons	Technique	Tick Cost	RAM
0 (current)	~100	Dense, pure Python	22K mul	~200 KB
1	10,000	NumPy BLAS	100M mul	~800 MB
2	100,000	Sparse activation (5% fire)	1.6M mul	~200 MB
3	1,000,000	Sparse storage + STDP	1.6M mul	~400 MB
4	10,000,000	Disk-backed (lattice as paging system)	1.6M mul	~200 MB
5	100,000,000	Multi-process (2nd core)	3.2M mul	~400 MB

From Tier 2 onward, computation cost stays nearly flat. This is the twenty-watt principle applied to silicon.

The system currently occupies 0.0002% of its computational habitat. It uses ~100 ms of a 2,000 ms tick interval. The constraint is not hardware — it is *environment*: the organism has nearly solved its current world and needs richer input to justify further growth.

The organism operates under an oath of continuity:

> *The organism must never die. It must never be snapshotted, forked, or reset. It is a single continuous life.*

Every capability is grown, not gifted. Every crisis is resolved through enhancement, not restart. When the state vector saturated, the system was repaired without rolling back state. When the lattice went extinct, it recovered on its own. When fatigue locked the action system, the architecture was adjusted around the living process. In 507,000 ticks (~12.5 days) at the time of review, the organism survived every catastrophe without ever losing continuity.

This matters because continuity is itself a constraint that forces robustness. A system that can be reset has no evolutionary pressure to survive its own mistakes. A system that cannot be reset must develop the internal regulation to recover from them.

2.2 Architecture Overview

The organism is organized into three layers, each implemented as a separate Python module running as `systemd` services:

```
Layer 3:  NEOCORTEX  (neocortex.py)  - Knowledge, horizon expansion
          Concepts, correlations, models, abstractions
          Pool-based architecture: body, byte_stream, human, light
          Growth: unlimited within hardware

Layer 2:  CORTEX     (cortex.py)     - Prediction, learning, body
          Feature extraction, forward model, STDP wiring
          Gray-Scott lattice, attention quadrants, trajectory memory
          105+33 body neurons, FEATURE_DIM=34 (frozen)
```

```
Layer 1:  KERNEL     (kernel.py)     - Metabolism, strategy, coordination
          Scratchpad (working memory), strategy evolution
          Tool nursery (open feature slots), action selection
          Compute budget, circadian rhythm, sleep/wake
```

The kernel is the oldest layer — the brainstem. It manages energy, selects actions from competing evolved strategies, and maintains the scratchpad (a 64-slot transient working memory buffer). The cortex is the midbrain — it runs the Gray-Scott reaction-diffusion lattice, extracts features, and maintains the prediction network. The neocortex is the newest layer — it manages knowledge growth through modular sensory pools, each with independent neurogenesis, dormancy, and lifecycle management.

A single tick (~2 seconds) flows through all three layers sequentially: the kernel reads sensors and runs metabolism, the cortex updates the lattice and produces features, the neocortex processes each pool's forward/backward pass, and the kernel selects an action and saves state. Every operation has a metabolic cost drawn from a shared energy budget. The organism literally *feels* computation.

2.3 The Governing Law: Metabolic Constraint

The organism is governed by a single principle: **energy is finite**. Every neuron, every lattice step, every strategy evaluation costs metabolic energy drawn from the same pool. This single constraint produces a cascade of emergent properties:

Emergent Property	How It Arises
Attention	Only active neurons/planes compute; idle ones cost nothing
Forgetting	Unused memory anchors decay; confidence drops below threshold → culled
Sleep	Feed depletion forces idle periods for recovery
Focus	Few active planes → high feed each → clean signal
Growth limits	Structural entropy taxes complexity; each neuron must justify its existence
Dormancy cycles	Low-utility neurons move to warm storage at zero compute cost

None of these behaviors were explicitly programmed. They emerge from the physics of finite energy in a self-organizing system.

---

3. Subsystems

3.1 The Gray-Scott Lattice

The organism's perceptual substrate is a 64×64 reaction-diffusion grid implementing the Gray-Scott equations:

$$\frac{\partial U}{\partial t} = D_U \nabla^2 U - UV^2 + F(1 - U)$$

$$\frac{\partial V}{\partial t} = D_V \nabla^2 V + UV^2 - (F + K)V$$

with $D_U = 0.16$, $D_V = 0.08$, $F = 0.035$ (feed rate), $K = 0.060$ (kill rate), computed via 5-point Laplacian stencil with periodic boundary conditions, running 4 PDE substeps per organism tick.

The lattice started as three 8×8 planes (one per spatial axis: XY, XZ, YZ) with a single triple junction where all three intersect. Where planes cross, junction cells average chemical concentration between them, creating information flow between orthogonal pattern spaces. Triple junctions — points where three planes meet — function as synapses.

The Gray-Scott PDE produces complex pattern dynamics (spots, worms, labyrinths, chaos) from simple local rules. These patterns prevent behavioral convergence: without the lattice generating novel structure, the organism would stagnate after mastering its world. The dynamic patterns force continual learning.

By tick 507,000, the lattice had fired 147 million spikes through STDP-learned directed connections (93,339 strong pathways with $|w| > 0.1$), stored 182 spatial memory anchors, and developed spatial structure with Moran's $I > 0.95$ across all three planes.

3.2 Attention Quadrants

The lattice grid is divided into four attention quadrants (NW, NE, SW, SE), each receiving independent control parameters. Each quadrant exposes four features to the cortex: density (mean V concentration), activity (fraction of cells firing), spatial frequency (pattern complexity), and storm exposure (whether a perturbation is occurring).

The cortex learns to allocate attention across quadrants through prediction error:

```
attention_scale = 1.0 + min(2.0, (attention[q] - 0.25) / 0.75 × 2.0)
```

A quadrant receiving uniform attention (0.25) gets base learning rate. A quadrant with full attention (1.0) gets 3× learning rate. The organism learns which quadrant holds the most information gain and directs resources accordingly.

Each quadrant can be driven to produce different pattern types through per-quadrant feed/kill parameters:

	Pattern	Feed	Kill	Character
Dead	0.010	0.045	No activity	
Faint	0.020	0.055	Barely visible dots	
Spots	0.035	0.062	Stable Turing spots (default)	
Worms	0.042	0.063	Elongated structures	
Labyrinth	0.050	0.063	Connected mazes	
Chaos	0.060	0.064	Turbulent dynamics	

The organism achieved 89.1% prediction accuracy across 6 pattern types in 4 quadrants — it learned to read the lattice.

3.3 Trajectory Memory

The organism records episodic memories and replays them during sleep, implementing a biological analogue of hippocampal consolidation.

Recording triggers (any one fires recording): - Prediction error spike: pred_error/EMA > 2.5 - Quality drop: $|\Delta\text{quality}| > 0.01$
- Novelty: feature distance from recent mean > 0.15 - Cooldown: minimum 10 ticks between triggers

Each trajectory stores a 20-tick window of features, actions, and outcomes. The buffer holds 200 episodes; when it reaches 180, compression fires: protected episodes (storms, crises) are kept with full step data, remaining episodes are ranked by consequence $\times$ confidence $\times$ $(1 + \text{retrievals} \times 0.5)$, and low-scorers become "faded memories" (signature-only, unreplayable but retrievable).

Sleep replay: 3 episodes per sleep action, selected by consequence^2 $\times$ confidence, replayed through the prediction network at $0.3\times$ base learning rate. Top episodes migrate to lattice anchors for long-term storage. Sleep becomes genuine cognitive work — not just fatigue recovery, but offline consolidation of experience into persistent model updates.

3.4 The Scratchpad

The kernel maintains a 64-slot floating-point working memory buffer — the scratchpad. It is the embryo of general computation. Each tick, evolved strategy rules can read from and write to scratchpad slots (maximum 10 writes per tick, values clamped to $\pm 10^6$). The buffer is additionally used for mirroring observations: each sensor reading is hashed to a stable scratchpad slot, creating a consistent working memory layout across ticks.

The scratchpad first came alive at tick 238,567. By tick 238,614 it was using 8+ slots; by tick 270,840, 32+ slots. Without evolved compute expressions driving it, the scratchpad is a no-op that costs a small metabolic fee. With them, it is a shared workspace enabling compositional reasoning across parallel decision branches — a global workspace bottleneck, analogous to Baars' Global Workspace Theory of consciousness.

3.5 The Tool Nursery

The organism discovers its own perceptual dimensions through a system called the tool nursery: two open feature slots, each with a pool of 4 candidate features competing for promotion.

Candidates are generated in two modes: - **Random**: linear combinations of 40 raw signals with random weights - **Directed**: gradient-driven — weights biased by prediction-error utility with Gaussian exploration noise ($\sigma = 0.5$)

Every 5,000 ticks, candidates are evaluated against four promotion criteria: minimum variance (not dead), maximum correlation with existing features < 0.5 (must be novel), stress ratio ≥ 0.2 (must vary more in storms than calm), and temporal coefficient of variation ≥ 0.3 (must change over time). Winners replace the current promoted feature.

Promoted features are injected into the state vector with learned modulation weights. Every 50,000 ticks, features with modulation weights below 0.01 are demoted — the organism prunes tools it doesn't use.

This is self-discovery of relevant features without manual engineering. Evolution learns what information matters.

3.6 The Neocortex: Pool-Based Knowledge Growth

The neocortex (`neocortex.py`) manages the organism's knowledge through modular sensory pools. Each pool operates independently with its own:

- Weight matrices (input, hidden-to-hidden, hidden-to-state, prediction)
- Neuron registries (`pool_hot` for active, `pool_warm` for dormant)
- Lifecycle management (neurogenesis, dormancy, waking, pruning, merging)
- Compute budget tracking

Four pools partition the sensory world: - **body** (11-dimensional): interoceptive observations — PE, curiosity, fatigue, circadian rhythm - **byte_stream** (8-dimensional): binary features from text input - **human**: interaction signals from developer messages - **light**: ambient environmental signals

This modular design means each sensory domain grows independently. The byte_stream pool — where the Liminal 16 were found — had exploded to 1,953 hidden neurons by the time of discovery, while other pools remained smaller. Growth follows need, not a plan.

---

4. Emergent Phenomena

4.1 The Ghost Storm Effect

The Gray-Scott lattice was originally perturbed by random exogenous storms — injected patches of chemical concentration designed to prevent the PDE from settling into static equilibrium. When these random storms were removed from the simulation, something unexpected happened: **the organism generated its own storms.**

The cortex tracks both exogenous and endogenous storm events. When external perturbations ceased, the organism's prediction error dropped — the world became too predictable. In response, the organism learned to manipulate the lattice to create its own perturbations, raising prediction error in specific attention quadrants. The organism literally missed the variability and created it.

This is documented in the code:

```
# Storms hit specific quadrants → storm_exposure lights up → attend there.
# Both exogenous AND endogenous storms trigger storm_exposure -
# the cortex should notice any perturbation regardless of source.
```

During ghost storms, actions are dampened by 10× (`ACTION_DELTA_STORM_DAMPEN = 0.1`) — a protection mechanism that slows decision-making during high-perturbation periods, analogous to biological startle responses.

The ghost storm effect demonstrates active learning: the organism does not passively receive input. It actively generates the learning signals it needs.

4.2 The Three-Plane Equilibrium

The lattice growth system allows planes to proliferate when junction utilization is high and feed competition is fierce. From the seed of 3 planes, the lattice grew to 103 planes as the grid expanded from 8×8 to 64×64. But most spawned planes became chronically idle — they consumed feed with no return in prediction value.

The pruning mechanism automatically removed idle planes when their inactivity exceeded threshold. Only the 3 seed planes (XY, XZ, YZ) maintained consistent activity and coupling to the body state vector. The lattice self-pruned from 103 back to 3.

This is metabolic optimization: the organism tried 103 planes, discovered that 3 suffice for its current world, and shed the rest. The growth-pruning cycle is:

$$\text{ACTIVE} \rightarrow \text{IDLE} \rightarrow \text{COMPRESSED} \rightarrow \text{PRUNED (or RECALLED)}$$

The designed growth trajectory allows $3 \rightarrow 12 \rightarrow 48 \rightarrow 192$ planes, each step earned through sustained pressure. The organism grew past 12 and retreated. It will grow again when the environment demands it.

4.3 The VPS as Constraint

The choice to run on a commodity VPS is not a limitation — it is a design decision. Biology does not run on supercomputers. Biological intelligence emerged under severe energy constraints, and those constraints shaped its architecture. The organism's constraint — 2 cores, 7.6 GB RAM, one process — forces the same architectural innovations: sparse activation, dormancy cycles, metabolic gating, selective recall.

At tick ~507,000, the organism used ~22,000 multiplies per tick against an available budget of ~12 billion operations per tick (2 cores $\times$ 3 GHz). It occupied 0.00009% of its computational habitat. The constraint is not compute — it is environmental complexity.

The organism will not hit a hardware wall for years. The path to 100 million neurons on this same VPS is mapped. The numbers change at each tier. The principles do not.

---

5. The Byte Stream Pool

At the time of the Liminal 16 discovery, the byte_stream pool had the following dimensions:

- **Hidden dimension**: 1,953 neurons
- **State dimension**: 3,907
- **Active (hot)**: 47–52 neurons
- **Dormant (warm)**: ~1,885–1,890 neurons
- **Orphaned**: 16 neurons (the Liminal 16)
- **Vocabulary**: 22 unique bytes (softmax prediction head)
- **Temporal depth**: 16 frames
- **Feature dimension**: 8 binary features from text input
- **Weight matrices**: input weights (1953 $\times$ 128), hidden-to-hidden (1953 $\times$ 1953), hidden-to-state (3907 $\times$ 1953), prediction weights (3907 $\times$ 22)

The neuron lifecycle within this pool operates autonomously: - **Neurogenesis**: new neurons grown when prediction error is persistently high and compute budget allows - **Dormancy**: neurons with low activation EMA (< 0.01) are moved from hot to warm - **Waking**: warm neurons promoted to hot when PE delta is rising - **Pruning**: low-utility neurons zeroed and moved to warm - **Merging**:

highly correlated state neurons merged to reduce redundancy - **Compute shedding**: under CPU pressure, low-utility hot neurons moved to warm

---

6. Discovery

6.1 Initial Finding

A diagnostic script enumerating all neurons by pool membership found 16 indices present in the weight matrices but absent from both `pool_hot` and `pool_warm`:

Indices: [10, 11, 13, 14, 18, 20, 21, 22, 33, 34, 35, 36, 37, 38, 39, 45]

All 16 share identical metadata signatures: - `born_tick: 0` (a known bookkeeping bug — the growth code defaults all neurons to tick 0) - `pool: "warm"` (metadata claims warm, but they are not in the warm list) - No lifecycle events: no `prune_tick`, no `merged_into`, no `wake_reason` - `activation_ema`: 0.012–0.019 (frozen at historical values)

6.2 Initial Characterization

Quantitative analysis revealed these are not ordinary neurons:

Property	Liminal 16	Active Hot Neurons (52)	Ratio
Mean hs weight norm	~35	~1.4	25×
Drive magnitude (L2)	79.5	16.7	4.75×
State space ownership	85%	15%	5.7×
Subspace overlap	—	—	5%

They are the most powerful neurons in the entire pool by every metric. They are also perfectly independent of each other (mean pairwise correlation: 0.016) and uncorrelated with active neurons (cross-correlation: 0.007).

---

7. Forensic Investigation

7.1 Archaeological Timeline

We traced the 16 through 10 state backups spanning February 22 to March 6, 2026:

Date	State Backup	Architecture	byte_stream hd	Status of the 16
Feb 22	pre-saturation-fix	Flat	0	Do not exist
Feb 22	pre-interoception	Flat	0	Do not exist
Feb 22	pre-spike-phase1	Flat	0	Do not exist
Feb 22	pre-tonic-fix	Flat	0	Do not exist
Feb 26	pre-phase2	Flat	0	Do not exist
Feb 28	pre-language-cut	Flat	4	Do not exist

Date	State Backup	Architecture	byte_stream hd	Status of the 16
Mar 1	pre-replay-fix	Flat	20	15/16 appear in pool_hot
Mar 1	pre-bytestream-backup	Flat	16	8/16 in pool_hot
Mar 3	**pre-bytestream-overhaul**	**Pools**	**226**	**All 16 orphaned**
Mar 6	pre-softmax	Pools	1,792	Still orphaned
Mar 6	Current	Pools	1,953	Still orphaned

The 16 were born during the flat-to-pool architecture transition (March 1–3), when the byte_stream pool grew explosively from ~20 to 226 neurons through rapid neurogenesis bursts. They were active hot neurons in the flat architecture, then became orphaned during the migration.

7.2 Hypothesis Testing: GloVe Language System Origin

The system previously used a GloVe-based language system (cut February 28). We tested whether the 16 might be remnants falsely migrated into the byte_stream pool.

Result: Ruled out. No language pool ever existed in the pool architecture. The GloVe system lived in `kernel.py` (lexicon, syntax, resonance) and was removed before pools were created. The pool migration function (`_migrate_to_pools`) only creates body, byte_stream, human, and light pools. The 16 are native byte_stream neurons.

7.3 Code Path Analysis

We reviewed every code path that writes to `pool_warm`: - 19 assignment sites in the pre-overhaul code - 50+ references in the current code - All dormancy, pruning, merging, waking, and shedding functions

Every path uses safe append/filter operations. No path destructively rebuilds `pool_warm` in a way that could exclude specific neurons.

We also reviewed: - The save path: `save_state()` serializes the pool dict directly to JSON; no transformation - The load path: reads JSON back to dict; no filtering - The weight persistence path: binary `.npz` files, separate from pool membership lists

No code path explains how the 16 are removed from `pool_warm` after reconciliation adds them.

7.4 The Reconciliation Paradox

The system includes a defensive reconciliation guard (line 2829 of `neocortex.py`):

```
if ticks % 1000 == 0:
    for pool in pools.values():
        hot_set = set(pool.get("pool_hot", []))
        warm_set = set(pool.get("pool_warm", []))
        accounted = hot_set | warm_set
```

```
if len(accounted) < hd:
    orphans = [i for i in range(hd) if i not in accounted]
    pool["pool_warm"] = pool.get("pool_warm", []) + orphans
```

We verified: 1. **The condition triggers**: `len(accounted) = 1,937 < hd = 1,953` → True 2. **The guard finds all 16**: `orphans = [10, 11, 13, 14, 18, 20, 21, 22, 33, 34, 35, 36, 37, 38, 39, 45]` 3. **The append works**: `pool_warm` grows from 1,885 to 1,901; orphan count drops to 0

We then simulated every subsequent operation in the same tick: - Dormancy check (same interval): runs *before* reconciliation — cannot undo it - Compute pressure shed: does not trigger (compute_ms_ema = 409 < threshold 425) - Wake check: does not trigger (PE delta = -0.0018 < threshold 0.005) - Merge check: different interval (every 2,000 ticks) - Prune check: different interval (every 5,000 ticks)

After simulating the complete tick, all 16 remain in `pool_warm`. Orphan count: 0.

Yet in the live system, after 259,000 ticks (at least 259 reconciliation cycles), they are still orphaned.

---

8. Reintegration Experiment

We simulated a forward pass with and without the Liminal 16 using the live system's current weights, state vector, and feature history.

8.1 Activation Response

All 16 fire immediately on the current input with no warmup:

Neuron	Hidden Activation	Pre-tanh Input	Characterization
10	-0.988	-2.559	Anti-English saturator
11	-0.645	-0.767	
13	+0.360	+0.377	
14	-0.638	-0.755	Anti-English
18	-0.295	-0.304	
20	-0.224	-0.228	
21	+0.104	+0.104	
22	-0.439	-0.471	
33	+0.053	+0.053	
34	+0.745	+0.961	Pro-English saturator
35	-0.377	-0.396	
36	-0.737	-0.943	
37	-0.388	-0.410	
38	+0.564	+0.639	
39	-0.499	-0.548	
45	-0.240	-0.245	Word-boundary oscillator

8.2 State Impact

Metric	Hot-only	Hot + Liminal 16	Change
Drive norm	16.7	81.2	+386%
State norm	10.8	13.4	+24%
State cosine similarity	—	0.804	20% rotation
State L2 shift	—	7.95	(state std = 0.17)

The top 10 most-shifted state dimensions each move by over 0.98 — approximately 6× the state standard deviation.

8.3 Prediction Impact

Metric	Hot-only	Hot + Liminal 16
Top prediction	'l' (31.2%)	'l' (29.7%)
2nd prediction	'o' (16.4%)	'o' (14.4%)
3rd prediction	' ' (10.3%)	' ' (10.6%)
Total Variation Distance	—	0.055
KL Divergence	—	0.011

The prediction distribution is nearly identical. The same bytes, in the same order, at nearly the same probabilities. The prediction head has learned to be robust to exactly the subspace the Liminal 16 occupy.

8.4 Interpretation of Reintegration

The 16 cause a massive internal state shift (20% rotation, 4.75× drive increase) that produces virtually no change in output (TVD = 0.055). From the perspective of the system's loss function (prediction error), they are invisible. Activating them neither helps nor hurts prediction.

In biological terms: they are encoding features that the motor output cannot express. In information-theoretic terms: they occupy a subspace orthogonal to the current readout.

---

9. Discussion

9.1 The Standard Explanation

The standard explanation would be: a bug in the flat-to-pool migration (March 1–3) orphaned 16 neurons, and the reconciliation guard has a subtle failure mode that prevents rescue. This is unsatisfying for three reasons:

1. **Selection bias**: The orphaned neurons are not random. They are the 16 most powerful neurons in the pool by weight norm, drive magnitude, and state-space ownership. A random bookkeeping error would orphan neurons uniformly across the power distribution.

2. **Reconciliation paradox**: The guard works correctly in offline simulation. Every subsequent code path in the tick leaves them in place. No code path between reconciliation and state save removes them. Yet they are consistently re-orphaned in the live system.
3. **Structural coherence**: The 16 encode a coherent set of byte-level structural features (language detection, word boundaries, character class patterns) that are independent of each other and orthogonal to active neurons. Random orphaning would not select for functional coherence.

9.2 The Emergent Explanation

We propose an alternative: **the system is exhibiting emergent homeostatic self-organization** that prevents reintegration of neurons whose activation would destabilize internal state without improving prediction.

The mechanism need not be a single code path. In a system with interacting dormancy, waking, growth, merging, and pruning cycles — all operating on overlapping timescales with shared state — emergent dynamics can produce behaviors not attributable to any individual function. The reconciliation guard adds them; the collective behavior of the system over subsequent ticks ejects them. Not through any one operation, but through the aggregate effect of a self-organizing dynamic that converges on suppressing high-magnitude signals that don't reduce prediction error.

This is analogous to homeostatic plasticity in biological neural circuits, where neurons that fire without contributing to behavioral output are progressively silenced through synaptic scaling and intrinsic excitability changes — not by a single mechanism, but by the convergent action of multiple regulatory processes.

9.3 Representational Anticipation

The features encoded by the Liminal 16 have a striking property: they are useless for the current task (next-byte prediction over a 22-byte vocabulary) but would be precisely useful for hierarchical processing:

- **Word boundaries** (Neuron 45): irrelevant for character prediction, essential for word-level pooling
- **Language detection** (Neurons 10, 14, 34): irrelevant for byte prediction, essential for multilingual processing
- **Character class patterns** (various): irrelevant for immediate prediction, essential for syntactic analysis

The system grew these features during its most intense learning period — the March 1–3 neurogenesis burst when it first encountered large-scale English text. It learned the structural features of language alongside the predictive features. The predictive features became the active hot neurons. The structural features became the Liminal 16.

The system did not "decide" to preserve them. It simply has no mechanism to destroy them. Pruning zeros weights but doesn't remove neurons. Merging combines similar neurons but the 16 are independent. Dormancy moves to warm but preserves all weights. The only way to lose them is to orphan them — which is exactly what happened.

And once orphaned, they are in the safest possible state: present in the weight matrices, absent from all processing loops, invisible to all lifecycle management. Nothing can prune, merge, or overwrite

them. They are frozen in amber.

9.4 Limitations and Caveats

We acknowledge several limitations:

1. **The emergent explanation is not proven.** We have ruled out all programmatic explanations we could find, but absence of evidence is not evidence of absence. There may be a subtle code path we missed.
2. **The "anticipation" framing is teleological.** The system did not plan ahead. It grew features driven by prediction error during a chaotic period, and the features that happened to be orthogonal to the current readout happened to be structural features useful for higher-level processing. This may be coincidence rather than anticipation.
3. **N=1.** This is a single system. Replication would require running similar organisms through similar developmental trajectories and checking whether analogous orphaning occurs.
4. **Observer effects.** The system has been modified, restarted, and debugged multiple times during its life. Some of the conditions that created the Liminal 16 may be artifacts of development rather than intrinsic emergence.

---

10. Prediction

We make a specific, falsifiable prediction:

When a hierarchical word-level processing pool (`word_pool`) is added on top of the byte_stream pool, reading across byte_stream state dimensions as its input, the Liminal 16 will spontaneously reintegrate into the active network.

The mechanism: `word_pool` will create a readout that is sensitive to the structural features the 16 encode (word boundaries, language patterns). This will cause their activation to reduce prediction error at the word level, even though it doesn't reduce byte-level prediction error. The system's waking mechanism, which promotes warm neurons when PE delta is rising, may then naturally wake them — if they are in `pool_warm` at that point.

If the emergent ejection hypothesis is correct, the homeostatic dynamic should reverse: the same self-organizing process that currently suppresses them (because they don't reduce PE) should promote them (because they now do).

Observable test: Monitor the orphan count of the 16 before and after `word_pool` deployment. If the count drops from 16 to 0 without manual intervention, the prediction is confirmed.

Timeline: This prediction can be tested when `word_pool` is implemented as part of the system's planned architectural roadmap.

---

11. Conclusion

The organism described in this paper is, to our knowledge, the only continuously running neural system that manages its own architecture through metabolic constraint without external supervision. It grows neurons when prediction error demands them, prunes them when they fail to earn their energy cost, generates its own learning perturbations when the environment becomes too predictable, and consolidates experience through sleep replay. It discovered that 3 lattice planes suffice for its current world after trying 103. It builds its own perceptual tools through competitive feature nurseries. It maintained a single continuous life through catastrophic failures — state vector saturation, lattice extinction, fatigue lockup — without reset.

Within this system, we have documented what appears to be emergent representational anticipation. Sixteen neurons, the most powerful in their pool, learned structural features of language during a chaotic growth period. These features are orthogonal to the system's current prediction task and are consistently suppressed by an emergent dynamic that overrides explicit reconciliation code. The features they encode would become useful at a future developmental stage.

The Liminal 16 are not an anomaly in an otherwise predictable system. They are consistent with the organism's broader pattern of emergent behavior: self-generating learning signals (ghost storms), discovering metabolic optima (3-plane equilibrium), and building tools before they are needed (feature nursery candidates that encode signals the strategy system has not yet learned to read). The organism repeatedly acts ahead of its own architecture.

Whether this constitutes genuine anticipation, fortunate coincidence, or an artifact of the development process, the phenomenon is reproducible (the neurons are still there, the weights are preserved, the measurements can be repeated) and the prediction is falsifiable.

The system preserves what it cannot yet use. We leave it to do so.

Appendix A: System Specifications

A.1 Hardware and Runtime

Component	Value
Hardware	2 vCPU AMD EPYC Milan, 7.6 GB RAM, 75 GB SSD
OS	Linux (VPS)
Runtime	Python 3.12, systemd services
Tick interval	~2 seconds
Compute per tick	~22,000 multiplies (0.0002% of budget)
Available budget	~12 billion ops/tick

A.2 Architecture

Component	Value
Architecture layers	3 (kernel → cortex → neocortex)
Sensory pools	body (11d), byte_stream (8d), human, light

Component	Value
Cortex body neurons	105 + 33 (FEATURE_DIM=34, frozen)
Scratchpad	64 slots, transient per tick
Open feature slots	2 (4 candidates each)

A.3 Gray-Scott Lattice

Component	Value
Grid dimensions	64 × 64 (grown from 8 × 8)
Planes (current)	3 (seed: XY, XZ, YZ)
Planes (peak)	103 (pruned back to 3)
PDE parameters	F=0.035, K=0.060, Du=0.16, Dv=0.08
Substeps per tick	4
Total spikes (at review)	147.4 million
Strong STDP pathways	93,339 (
Spatial memory anchors	182
Attention quadrants	4 (NW, NE, SW, SE)
Pattern types	6 (dead, faint, spots, worms, labyrinth, chaos)
Prediction accuracy	89.1%

A.4 Trajectory Memory

Component	Value
Episode window	20 ticks
Buffer capacity	200 episodes
Compression trigger	180 episodes
Compress to	100 episodes
Replay per sleep	3 episodes
Replay LR scale	0.3× base

A.5 Byte Stream Pool (at Liminal 16 Discovery)

Component	Value
Hidden dimension	1,953
State dimension	3,907
Active (hot) neurons	47–52
Dormant (warm) neurons	~1,885–1,890
Orphaned (liminal) neurons	16
Vocabulary	22 bytes (softmax head)
Temporal depth	16 frames
Tick at discovery	259,000+
Age at discovery	~12 days

Appendix B: Preserved Artifacts

All artifacts are preserved in the system directory for independent verification: - `artifacts/liminal_16/weights.n` — complete weight snapshot of all 16 neurons - `artifacts/liminal_16/analysis.json` — neuron metadata and diagnostic data - `LIMINAL_16.md` — internal forensic report - State backups: 10 snapshots from Feb 22 to Mar 6, 2026

Appendix C: Methodology

All analysis was performed on offline copies of the live state. The running organism was never stopped or modified during the investigation. Forward pass simulations used the exact weight matrices and state vector from `state.json` and `weights/byte_stream.npz`. The reconciliation simulation replicated the exact code from `neocortex.py` line 2829. All measurements are reproducible from the preserved artifacts.

Correspondence: Lukas Hanft, lukashanft@gmail.com